In [1]:
```
# 1. Leap Year Checker
# Accept a year.
# First check whether it is divisible by 4.
# If yes, check whether it is divisible by 100.
# If it is, check whether it is divisible by 400.
# Display whether the year is a leap year.
year=int(input("enter the year"))
if(year%4==0):
    if(year%100==0):
        if(year%400==0):
            print("it is a leap year")
        else:
            print("it is not a leap year")
    else:
        print("it is a leap year")
else:
    print("it is not leap year")
```

```
it is a leap year
```

In [2]:
```
# 2. Triangle Classification
# Accept three side lengths.
# First check whether they can form a valid triangle.
# If valid, check whether all three sides are equal, exactly two sides are equal
# or all sides are different.
a=float(input("enter the first side:"))
b=float(input("enter the second side:"))
c=float(input("enter the third side:"))
if(a+b>c and a+c>b and b+c>a):
    if(a==b and b==c):
        print("Equilateral triangle")
    else:
        if(a==b or b==c or a==c):
            print("Isosceles triangle")
        else:
            print("Scalene triangle")
else:
    print("Invalid triangle")
```

```
Isosceles triangle
```

In [4]:
```
# 3. Employee Bonus Calculator
# Accept years of experience and performance rating.
# If experience is at least 5 years, check the rating.
# Give a 20% bonus for an excellent rating and 10% for a good rating.
# If experience is below 5 years, give a 5% bonus only for an excellent rating.
sal=float(input("enter the salary:"))
year=int(input("enter the year of experience:"))
rat=input("enter the good or excellent rating:")
if(year>=5):
    if(rat=="excellent"):
        print("you got 20% bonus")
        bonus=sal*20/100
    elif(rat=="good"):
        print("you got 10% bonus")
        bonus=sal*10/100
    else:
        print("no bonus")
elif(rat=="excellent"):
```

```
    print("you got 5% bonus")
    bonus=sal*5/100
else:
    print("no bonus")
    bonus=0
print(f"the bonus is:{bonus}\nThe total salary is:{sal+bonus}")
```

```
you got 10% bonus
the bonus is:5000.0
The total salary is:55000.0
```

In [5]:
```
# 4. Password Strength Checker
# Accept a password.
# If its length is at least 8 characters, check whether it contains a digit.
# If it does, check whether it contains an uppercase letter.
# Classify it as strong or moderate based on these conditions.
# Otherwise, classify it as weak.
p=input("enter the password")
plen=len(p)
if(plen>=8):
    if any(char.isdigit() for char in p):
        if any(char.isupper() for char in p):
            print("Strong password")
        else:
            print("moderate password")
    else:
        print("Weak password")
else:
    print("Weak password")
```

```
Strong password
```

In [6]:
```
# 5. Hospital Consultation Priority
# Accept a patient's temperature and oxygen saturation.
# If oxygen saturation is below 90%, classify as high priority.
# Otherwise, check whether the temperature is at least 39°C.
# If so, classify as urgent; otherwise, classify as normal priority.
# (Simplified programming exercise, not a medical triage tool.)

temp=float(input("enter the temperature in celsius:"))
oxygen=float(input("enter oxygen saturation:"))
if oxygen<90:
    print("High priority")
else:
    if temp>=39:
        print("Urgent")
    else:
        print("normal priority")
```

```
High priority
```

In [8]:
```
# 6. Income Tax Slab Calculator
# Accept annual taxable income. If income is up to ₹3 lakh, tax is 0%.
# Otherwise, check whether it is up to ₹7 lakh and apply a 5% rate to income abo
# For income above ₹7 lakh, apply progressive slabs of 0%, 5%, and 10% to the re
# Calculate the total tax.

income = float(input("Enter annual taxable income: "))

if income <= 300000:
    tax = 0
```

```
else:
    if income <= 700000:
        tax = (income - 300000) * 0.05
    else:
        tax = (400000 * 0.05) + ((income - 700000) * 0.10)

print("Total tax: Rs.", tax)
```

```
Total tax: Rs. 5000.0
```

In [9]:
```
# 7. College Admission
# Accept marks in Mathematics, Physics, and Chemistry.
# If Mathematics is at least 60, check Physics.
# If Physics is at least 50, check Chemistry.
# If Chemistry is at least 50 and the total is at least 180, display "Eligible";
# otherwise, display "Not Eligible".
maths = int(input("Enter Mathematics marks: "))
physics = int(input("Enter Physics marks: "))
chemistry = int(input("Enter Chemistry marks: "))

total = maths + physics + chemistry

if maths >= 60:
    if physics >= 50:
        if chemistry >= 50:
            if total >= 180:
                print("Eligible")
            else:
                print("Not Eligible")
        else:
            print("Not Eligible")
    else:
        print("Not Eligible")
else:
    print("Not Eligible")
```

```
Eligible
```

In [10]:
```
# 8. Online Food Delivery
# Accept order amount, distance, and whether the customer has a membership.
# If the order amount is at least ₹500, check whether the distance is within 5 k
# Apply free delivery if within 5 km; otherwise, charge ₹40.
# If the order is below ₹500, check membership status and calculate delivery cha

amount = float(input("Enter order amount: Rs. "))
distance = float(input("Enter delivery distance in km: "))
membership = input("Do you have membership? (yes/no): ").lower()

if amount >= 500:
    if distance <= 5:
        delivery = 0
    else:
        delivery = 40
else:
    if membership == "yes":
        delivery = 20
    else:
        delivery = 50
```

```
print("Delivery charge: Rs.", delivery)
print("Total amount: Rs.", amount + delivery)
```

```
Delivery charge: Rs. 20
Total amount: Rs. 420.0
```

In [11]:
```
# 9. Insurance Premium Category
# Accept age, smoking status, and annual income.
# If the applicant is under 30, check smoking status to determine the risk categ
# If age is 30 or above, check income and smoking status to classify the applica
# Define your own clear rules for each category.
age = int(input("Enter age: "))
smoking = input("Do you smoke? (yes/no): ").lower()
income = float(input("Enter annual income: Rs. "))

if age < 30:
    if smoking == "yes":
        print("Medium risk")
    else:
        print("Low risk")
else:
    if income >= 500000:
        if smoking == "yes":
            print("Medium risk")
        else:
            print("Low risk")
    else:
        if smoking == "yes":
            print("High risk")
        else:
            print("Medium risk")
```

```
Medium risk
```

In [12]:
```
# 10. Mini Banking System
# Accept account type, account balance, and transaction type (deposit or withdra
# If the account type is savings, check the transaction type.
# For withdrawals, check minimum balance requirements;
# for deposits, validate the deposit amount.
# If the account type is current, apply a different minimum balance rule.

account = input("Enter account type (savings/current): ").lower()
balance = float(input("Enter account balance: Rs. "))
transaction = input("Enter transaction type (deposit/withdrawal): ").lower()
amount = float(input("Enter transaction amount: Rs. "))

if account == "savings":
    if transaction == "deposit":
        if amount > 0:
            balance = balance + amount
            print("Deposit successful")
            print("New balance: Rs.", balance)
        else:
            print("Invalid deposit amount")
    else:
        if transaction == "withdrawal":
            if amount > 0:
                if balance - amount >= 1000:
                    balance = balance - amount
                    print("Withdrawal successful")
                    print("New balance: Rs.", balance)
```

```
                else:
                    print("Minimum balance of Rs. 1000 must be maintained")
            else:
                print("Invalid withdrawal amount")
        else:
            print("Invalid transaction type")

else:
    if account == "current":
        if transaction == "deposit":
            if amount > 0:
                balance = balance + amount
                print("Deposit successful")
                print("New balance: Rs.", balance)
            else:
                print("Invalid deposit amount")
        else:
            if transaction == "withdrawal":
                if amount > 0:
                    if balance - amount >= 5000:
                        balance = balance - amount
                        print("Withdrawal successful")
                        print("New balance: Rs.", balance)
                    else:
                        print("Minimum balance of Rs. 5000 must be maintained")
                else:
                    print("Invalid withdrawal amount")
            else:
                print("Invalid transaction type")
    else:
        print("Invalid account type")
```

```
Withdrawal successful
New balance: Rs. 10000.0
```

In [ ]: